Lecture 10

The finite well: a spectrum in two parts

Objectives. *A finite well has both bound levels and a continuum. For a real bounded potential that vanishes outside a bounded interval and has at most finitely many jumps, we prove self-adjointness, simple negative levels and essential spectrum* $[0,\infty)$*. We count the square-well levels, and prove that every such well with* $\int V < 0$ *binds.*

- Match a square-well bound state and count its levels.
- Distinguish bound eigenvectors from non-square-integrable solutions.
- Use a trial state to prove binding when $\int V < 0$.

Lecture 10, first part

Bound states and the square well

10.1 Adding a bounded potential

A potential changes the energy of a particle by an amount that depends on its position. We take potentials that are bounded and act only in a bounded region.

The square well is the model we aim at. Let $V_0 > 0$ and $w > 0$, and let $V = -V_0$ on $(-w, w)$ and $V = 0$ outside: a well of depth V_0 and half-width w. For an energy E in $(-V_0, 0)$, put $k = \sqrt{2m(E+V_0)}/\hbar$ inside the well and $\eta = \sqrt{-2mE}/\hbar$ outside.

We first find an even level as physics texts do, in a provisional calculation. Inside the well we take the solution $A\cos kx$ of $-(\hbar^2/2m)\psi'' + V\psi = E\psi$, and for $x > w$ the decaying solution $Be^{-\eta x}$. We join them so that ψ and ψ' are continuous at w:

$$A\cos kw = Be^{-\eta w}, \qquad -Ak\sin kw = -\eta Be^{-\eta w}.$$

If $\cos kw = 0$, then $B = 0$ and then $A = 0$, so there is no state. Otherwise, for a nonzero state, the second condition divided by the first gives $\eta = k\tan kw$.

This calculation assumes two things that we have not proved. It assumes that an eigenvector of the Hamiltonian is continuously differentiable, also where V jumps. It also assumes that a function joined in this way is an eigenvector. Before we solve the well, we prove both, for every such potential. The Hamiltonian is self-adjoint (this section). The differential equation finds exactly its eigenvectors (§ 10.2). Its levels are simple (§ 10.3). § 10.4 then counts the levels of the square well.

Definition 10.1 (Potential). A *potential* is a real function V on $\mathbb{R}$ that is bounded, that vanishes outside a bounded interval $[-R, R]$, and that is continuous except at finitely many points, where it has limits from both sides. Its *Hamiltonian* is $H = H_0 + V$ on $D(H) = D(H_0)$, where V also stands for multiplication by V.

Proposition 10.2 (A bounded potential keeps self-adjointness)**.** *For every potential V, the Hamiltonian $H = H_0 + V$ is self-adjoint.*

Proof. Multiplication by V maps every $\psi \in L^2(\mathbb{R})$ to $V\psi$, with $\|V\psi\| \le \sup|V|\,\|\psi\|$, and $\langle \varphi, V\psi\rangle = \langle V\varphi, \psi\rangle$ since V is real. So H is symmetric, as a sum of two symmetric operators on $D(H_0)$. Let $\varphi \in D(H^\dagger)$, with $\chi = H^\dagger\varphi$. For every $\psi \in D(H_0)$,

$$\langle \varphi, H_0\psi\rangle = \langle \varphi, H\psi\rangle - \langle \varphi, V\psi\rangle = \langle \chi - V\varphi, \psi\rangle\,.$$

By Definition 0.2, $\varphi \in D(H_0^\dagger)$, which is $D(H_0)$ since H_0 is self-adjoint (Theorem 4.6), and $H_0\varphi = \chi - V\varphi$. So $\varphi \in D(H)$ and $H\varphi = \chi$. Hence $H^\dagger = H$. ∎

The Kato–Rellich theorem (Teschl, Thm 6.4) extends this to some unbounded potentials, such as the Coulomb potential of Lecture 14. The lowest energy of H can be bounded from its means. The next proposition holds for every self-adjoint operator. We use it to bound the spectrum of H from below, and in § 10.6 to find a level from a single state.

Proposition 10.3 (The bottom of the spectrum)**.** *Let A be self-adjoint, and suppose that*

$$\lambda_0 = \inf\{\langle \psi, A\psi\rangle :\ \psi \in D(A),\ \|\psi\| = 1\}$$

is finite. Then $\sigma(A) \subseteq [\lambda_0, \infty)$, and $\lambda_0 \in \sigma(A)$.

Proof. The first claim says that no point of the spectrum lies below the infimum, and Weyl's criterion gives it directly. For the second claim we argue by contradiction: if λ_0 lay outside $\sigma(A)$, the means would stay above λ_0 by a fixed amount.

First, since $\sigma(A) \subseteq \mathbb{R}$ by the first claim of Theorem 6.3, let $\lambda < \lambda_0$ be real. For a unit vector $\psi \in D(A)$, by the Cauchy–Schwarz inequality, $\|(A-\lambda)\psi\| \ge \langle \psi, (A-\lambda)\psi\rangle \ge \lambda_0 - \lambda$. So λ has no Weyl sequence, and by Weyl's criterion it does not lie in $\sigma(A)$.

Next, put $B = A - \lambda_0$, so that $\langle \psi, B\psi\rangle \ge 0$ for every $\psi \in D(A)$, and suppose that $\lambda_0 \notin \sigma(A)$. Then $K = B^{-1}$ is bounded, not 0, and maps $\mathcal{H}$ onto $D(A)$. For $\varphi \in \mathcal{H}$ and $\psi = K\varphi$,

$$\langle \varphi, K\varphi\rangle = \langle B\psi, \psi\rangle = \langle \psi, B\psi\rangle \ge 0.$$

In the same way $\langle \varphi, K\varphi'\rangle = \langle K\varphi, \varphi'\rangle$, since B is symmetric. Let $\delta > 0$. Then $s(\varphi, \varphi') = \langle \varphi, K\varphi'\rangle + \delta\,\langle \varphi, \varphi'\rangle$ is an inner product on $\mathcal{H}$, since K is symmetric and $s(\varphi,\varphi) \ge \delta\,\|\varphi\|^2$. So the Cauchy–Schwarz inequality (*Linear Algebra*, Thm 7.12) holds for it. With $\varphi' = K\varphi$, and letting $\delta \to 0$, it gives

$$\|K\varphi\|^4 = \left|\left\langle \varphi, K^2\varphi\right\rangle\right|^2 \le \langle \varphi, K\varphi\rangle\,\left\langle K\varphi, K^2\varphi\right\rangle \le \langle \varphi, K\varphi\rangle\,\|K\|\,\|K\varphi\|^2\,.$$

Divide by $\|K\varphi\|^2$ when it is not 0, and put $\psi = K\varphi$, which runs through $D(A)$. Hence $\|\psi\|^2 \le \|K\|\,\langle \psi, B\psi\rangle$ for every $\psi \in D(A)$ (for $\psi = 0$ trivially), and the means satisfy $\langle \psi, A\psi\rangle \ge \lambda_0 + 1/\|K\|$ for unit ψ, against the definition of λ_0. ∎

So every unit vector $\psi \in D(A)$ gives an upper bound $\langle \psi, A\psi\rangle$ for the least point of $\sigma(A)$. This is the *variational principle*: the least point of $\sigma(A)$ is the infimum of the means. A unit vector used in this way is called a *trial state*. For the box Hamiltonian of Theorem 1.8, $\langle \psi, H\psi\rangle = \sum_n E_n\,|c_n|^2 \ge E_1$, with equality for u_1. For a potential, $\langle \psi, H_0\psi\rangle = \int (p^2/2m)\left|\tilde\psi(p)\right|^2\,\mathrm{d}p \ge 0$, so the least point of $\sigma(H)$ is at least $-\sup|V|$.

Self-check 10.1

1. Let $V = -V_0$ on $(-w, w)$ and $V = 0$ outside, with $V_0 > 0$. Show that $\sigma(H) \subseteq [-V_0, \infty)$.

10.2 The eigenvalue equation

The calculation of § 10.1 solved the differential equation $-(\hbar^2/2m)\psi'' + V\psi = E\psi$ on each piece and joined the pieces smoothly. Here we prove that this finds exactly the eigenvectors of H. The tool is the free resolvent of Proposition 6.11, extended to every vector. We need every vector, since the resolvent will act on a function built from an eigenvector, which at the start is known only to be square-integrable. Recall its kernel $G(x) = (m/\hbar^2\eta)e^{-\eta|x|}$ at $z = -\hbar^2\eta^2/2m$, with $\eta > 0$, and write $G_x(y) = G(x - y)$.

Lemma 10.4 (The free resolvent on every vector). *Let $\eta > 0$, $z = -\hbar^2\eta^2/2m$ and $\varphi \in L^2(\mathbb{R})$, and put*

$$u(x) = \langle G_x, \varphi \rangle = \int_{-\infty}^{\infty} G(x - y)\, \varphi(y)\, \mathrm{d}y.$$

1. *The function u is $(H_0 - z)^{-1}\varphi$. For all x and x',*

$$|u(x)| \le \|G\|\, \|\varphi\|, \qquad |u(x) - u(x')| \le \|G_x - G_{x'}\|\, \|\varphi\|,$$

   with $\|G\|^2 = m^2/\hbar^4\eta^3$ and $\|G_x - G_{x'}\|^2 = (2m^2/\hbar^4\eta^3)\big(1 - (1 + \eta h)e^{-\eta h}\big)$, $h = |x - x'|$. In particular u is continuous.
2. *If φ is continuous except at finitely many points, where it has limits from both sides, then u is continuously differentiable, and at every point where φ is continuous, u is twice differentiable with*

$$-\frac{\hbar^2}{2m}\, u'' - zu = \varphi.$$

Proof.

1. The two bounds are the Cauchy–Schwarz inequality, and the two norms follow from $\int e^{-\eta|y|}e^{-\eta|y-h|}\,\mathrm{d}y = (h + 1/\eta)e^{-\eta h}$ for $h \ge 0$. By the standing fact S4 there are smooth φ_n, each vanishing outside a bounded interval, with $\varphi_n \to \varphi$. By Proposition 6.11, $(H_0 - z)^{-1}\varphi_n$ is the function $u_n(x) = \langle G_x, \varphi_n \rangle$. Since $(H_0 - z)^{-1}$ is bounded, $u_n \to (H_0 - z)^{-1}\varphi$. Moreover $|u_n(x) - u(x)| \le \|G\|\, \|\varphi_n - \varphi\|$, so $u_n \to u$ uniformly, and so in norm on every bounded interval J. Hence $1_J u = 1_J (H_0 - z)^{-1}\varphi$ for every J, and $u = (H_0 - z)^{-1}\varphi$.
2. Split the integral at $y = x$, as in the proof of Proposition 6.11. The two integrals converge by the Cauchy–Schwarz inequality. At a point where φ is continuous, by the fundamental theorem of calculus, the two terms $\varphi(x)$ that come from the limits cancel in u'. At a jump of φ they cancel in the derivatives from the left and from the right, so these agree. Hence u' exists everywhere and is the continuous function of that proof, and differentiating it at a point where φ is continuous gives $u'' = \eta^2 u - (2m/\hbar^2)\varphi$. ∎

One fact about the differential equation is used in the next proposition, and again in § 10.3. A solution is fixed by its value and its slope at one point.

Lemma 10.5 (Uniqueness). *Let V be a potential, E real, and I an interval. Let ψ be continuously differentiable on I, twice differentiable where V is continuous, and satisfy there*

$$-\frac{\hbar^2}{2m}\psi'' + V\psi = E\psi. \tag{10.1}$$

If $\psi(x_0) = \psi'(x_0) = 0$ at one point $x_0 \in I$, then $\psi = 0$ on I.

Proof. Let $C = (2m/\hbar^2)\sup|V - E|$ and $y = |\psi|^2 + |\psi'|^2$. Where V is continuous, $|\psi''| \le C|\psi|$ by (10.1), so

$$|y'| = \left|2\operatorname{Re}(\overline{\psi}\psi') + 2\operatorname{Re}(\overline{\psi'}\psi'')\right| \le 2(1+C)\,|\psi|\,|\psi'| \le (1+C)\,y,$$

using $2|a|\,|b| \le |a|^2 + |b|^2$. So $e^{-(1+C)x}y$ is nonincreasing and $e^{(1+C)x}y$ is nondecreasing on I, by the mean value theorem on each piece. Since $y \ge 0$ and $y(x_0) = 0$, the first gives $y = 0$ for $x \ge x_0$ and the second for $x \le x_0$. ∎

By Lemma 10.5, two solutions of (10.1) on an interval I with the same value and slope at one point agree on I, since their difference is a solution. Where V is a constant c, the solutions on I are therefore the combinations of $\cos kx$ and $\sin kx$ if $E - c = \hbar^2k^2/2m > 0$, of 1 and x if $E = c$, and of $e^{\mu x}$ and $e^{-\mu x}$ if $E - c = -\hbar^2\mu^2/2m < 0$.

A class of functions contains at most one continuous function, since two continuous functions that differ at a point differ on an interval around it.

Proposition 10.6 (The eigenvalue equation). *Let V be a potential, E a real number and $\psi \in L^2(\mathbb{R})$. Then $\psi \in D(H)$ and $H\psi = E\psi$ if and only if ψ is (the class of) a continuously differentiable function that is twice differentiable at every point where V is continuous and satisfies there* (10.1).

Proof. Fix $\eta > 0$ and $z = -\hbar^2\eta^2/2m$. Suppose first that $H\psi = E\psi$. Then $(H_0 - z)\psi = \varphi$ with $\varphi = (E - z - V)\psi$, so $\psi = (H_0 - z)^{-1}\varphi$, which by the first claim of Lemma 10.4 is a continuous function. So φ is continuous except at the jumps of V. By its second claim, ψ is continuously differentiable, and $-(\hbar^2/2m)\psi'' - z\psi = (E - z - V)\psi$ where V is continuous. This is (10.1).

Conversely, let ψ be such a function, and put $\varphi = (E - z - V)\psi$. Let $u = (H_0 - z)^{-1}\varphi$. By the second claim of Lemma 10.4, u is continuously differentiable and $-(\hbar^2/2m)u'' - zu = \varphi$ where V is continuous. So $d = u - \psi$ is continuously differentiable, with $d'' = \eta^2 d$ except at finitely many points. By the mean value theorem, d' is then differentiable at those points too, with $d'' = \eta^2 d$ everywhere. So, by Lemma 10.5 with $V = 0$ and $E = z$, $d = ae^{\eta x} + be^{-\eta x}$, and since d is square integrable, $a = b = 0$. Hence $\psi = u \in D(H_0)$, and $(H_0 - z)\psi = (E - z - V)\psi$, that is, $H\psi = E\psi$. ∎

So the smooth joining is not an extra rule. A solution with a kink, or one that is not square integrable, is not an eigenvector.

Self-check 10.2

1. Show that $\psi(x) = e^{-\eta|x|}$ satisfies $-(\hbar^2/2m)\psi'' = z\psi$ at every $x \ne 0$, with $z = -\hbar^2\eta^2/2m$. Why is z not an eigenvalue of H_0 with the eigenvector ψ?

10.3 Bound states are simple

An eigenvector of H is a *bound state*, and its eigenvalue a *level* of H. Like the stationary waves of Lecture 1, a bound state has a definite energy, and, as we show now, it decays outside the well. We use Lemma 10.5 to prove that the levels are simple.

Theorem 10.7 (Bound states). *Let V be a potential that vanishes outside $[-R, R]$. Every eigenvalue E of H is negative and simple. With $\eta = \sqrt{-2mE}/\hbar$, every eigenvector is a multiple of $e^{-\eta x}$ for $x \geq R$ and of $e^{\eta x}$ for $x \leq -R$.*

Proof. Let $H\psi = E\psi$. By Proposition 10.6, ψ solves (10.1), and for $x > R$ this is the free equation. If $E = \hbar^2 k^2/2m > 0$, then $\psi = a\cos kx + b\sin kx$ there, and each period contributes $(\pi/k)(|a|^2 + |b|^2)$ to $\int |\psi|^2$. Since ψ is square integrable, $a = b = 0$. If $E = 0$, then $\psi = a + bx$, and again $a = b = 0$. In both cases ψ and ψ' vanish beyond R, and Lemma 10.5 gives $\psi = 0$. So $E < 0$, and $\psi = ae^{\eta x} + be^{-\eta x}$ for $x > R$, with $a = 0$ since ψ is square integrable. The same holds for $x < -R$.

Let ψ_1 and ψ_2 be eigenvectors for E, and let $W = \psi_1\psi_2' - \psi_1'\psi_2$. It is continuous, and where V is continuous (10.1) gives $W' = \psi_1\psi_2'' - \psi_1''\psi_2 = 0$. So W is constant, and for $x > R$, where both are multiples of $e^{-\eta x}$, $W = 0$. Choose x_0 with $\psi_1(x_0) \neq 0$ and put $c = \psi_2(x_0)/\psi_1(x_0)$. Then $\psi_2 - c\psi_1$ and, since $W(x_0) = 0$, its derivative vanish at x_0. By Lemma 10.5, $\psi_2 = c\psi_1$. ∎

The *Wronskian* W of the proof returns in Lecture 11. A bound state decays like $e^{-\eta|x|}$, so the probability of finding the particle at $|x| \geq d$, for $d \geq R$, is a multiple of $e^{-2\eta d}$. In *Quantum Mechanics I* a level could be degenerate. On the line, with a potential, it cannot.

Self-check 10.3

1. Let ψ be a bound state. Show that $\overline{\psi}$ is a bound state for the same level, and that $e^{i\theta}\psi$ is a real function for some real θ.

10.4 The square well

Take the square well of § 10.1, of depth V_0 and half-width w. Its strength is measured by the number

$$\gamma = \frac{w\sqrt{2mV_0}}{\hbar}.$$

For a level E in $(-V_0, 0)$, with k and η as in § 10.1, $k^2 + \eta^2 = 2mV_0/\hbar^2$, so the point $(kw, \eta w)$ lies on the circle $(kw)^2 + (\eta w)^2 = \gamma^2$. We write $\lceil t \rceil$ for the least integer $n \geq t$.

Theorem 10.8 (The square well). *The square well has exactly $N = \lceil 2\gamma/\pi \rceil$ levels $E_1 < E_2 < \cdots < E_N$, all in $(-V_0, 0)$. The bound state of E_j is even for odd j and odd for even j. For each j, the point $(k_j w, \eta_j w)$ is the one point of the circle with $(j-1)\pi/2 < k_j w < j\pi/2$ and*

$$\eta_j = k_j \tan(k_j w) \quad \text{for odd } j, \qquad \eta_j = -k_j \cot(k_j w) \quad \text{for even } j.$$

Figure 10.1 shows the crossings and their bound states for $\gamma = 5$.

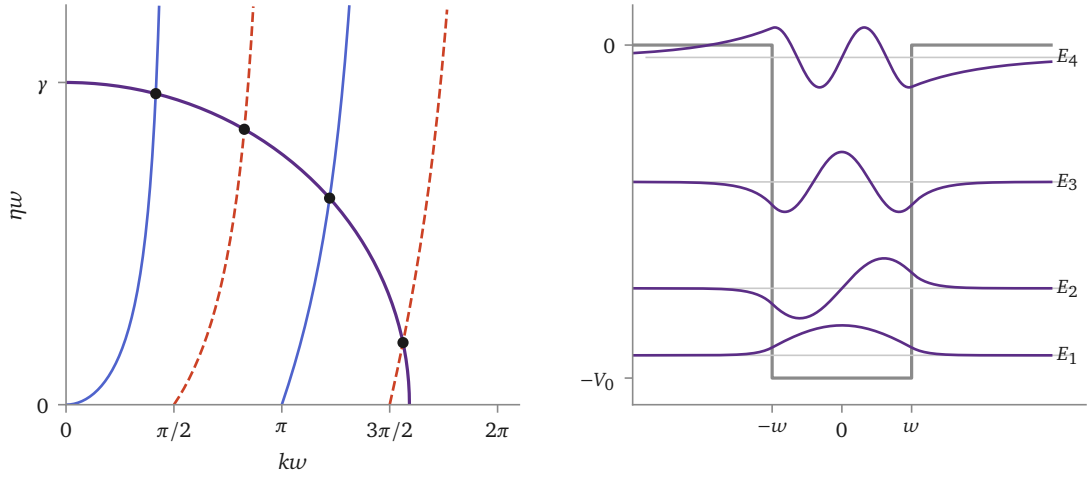


Figure 10.1: The square well with $\gamma = 5$. Left: the circle $(kw)^2 + (\eta w)^2 = \gamma^2$ crosses the even branches $\eta w = kw \tan kw$ (solid) and the odd branches $\eta w = -kw \cot kw$ (dashed) at $\lceil 10/\pi \rceil = 4$ points. Right: the four bound states, each drawn about the height of its level E_j; the vertical size of each wave is not to scale.

Proof.

1. *Parity.* Let ψ be a bound state for E. Since V is even, $\psi(-x)$ also solves (10.1), so by Proposition 10.6 and Theorem 10.7 $\psi(-x) = c\psi(x)$. Applying this twice gives $c^2 = 1$, so ψ is even or odd. An even ψ has $\psi'(0) = 0$, and an odd ψ has $\psi(0) = 0$.
2. *No level at or below $-V_0$.* By Theorem 10.7, for $x \geq w$ the bound state is $Be^{-\eta x}$, with $B \neq 0$ by Lemma 10.5, so $\psi'(w)/\psi(w) = -\eta < 0$. If $E \leq -V_0$, then inside the well an even ψ is a multiple of $\cosh \mu x$ or of 1, and an odd one of $\sinh \mu x$ or of x, with $\mu \geq 0$. Each has $\psi'(w)/\psi(w) \geq 0$. Since ψ and ψ' are continuous at w, this is impossible.
3. *Matching.* For $-V_0 < E < 0$, an even bound state is $A \cos kx$ inside, and $Be^{-\eta x}$ for $x \geq w$ by Theorem 10.7. By Proposition 10.6, ψ and ψ' are continuous at w, so the two conditions of the provisional calculation in § 10.1 hold. As there, $\cos kw \neq 0$ and $\eta = k \tan kw$. An odd bound state is $A \sin kx$ inside, and in the same way $\eta = -k \cot kw$. Conversely, if $k, \eta > 0$ satisfy one of these conditions and $k^2 + \eta^2 = 2mV_0/\hbar^2$, the function so defined is continuously differentiable, square integrable and solves (10.1) off $\pm w$. By Proposition 10.6, it is a bound state.
4. *Counting.* Since $\eta > 0$, the even condition needs $\tan kw > 0$, and the odd one $\cot kw < 0$. So kw lies in an interval $((j-1)\pi/2, j\pi/2)$, with j odd for even states and even for odd ones. On that interval the function $u \tan u$ or $-u \cot u$, with $u = kw$, rises from 0 to ∞, since its derivative $\tan u + u/\cos^2 u$ or $-\cot u + u/\sin^2 u$ is positive there, while the circle's height $\sqrt{\gamma^2 - u^2}$ falls. They cross exactly once if $(j-1)\pi/2 < \gamma$, and not at all otherwise. The integers $j \geq 1$ with $j - 1 < 2\gamma/\pi$ are $1, \dots, \lceil 2\gamma/\pi \rceil$. As j grows, $k_j w$ grows and $\eta_j w$ falls, so $E_j = -\hbar^2 \eta_j^2 / 2m$ rises. ∎

Every square well, however shallow, has at least one level, with an even bound state. A deeper or wider well has at least as many, and one more each time γ passes a multiple of $\pi/2$.

At $\gamma = n\pi/2$, for an integer $n \geq 1$, the circle meets the next branch at $\eta w = 0$, that is, at $E = 0$. There the function $\cos kx$ (for even n) or $\sin kx$ (for odd n) inside, continued by its

value at w for $x > w$ and by its value at $-w$ for $x < -w$, is continuously differentiable and solves (10.1) with $E = 0$. It does not decay, so it is not square integrable and not a bound state, as Theorem 10.7 requires. A new level appears only when γ passes $n\pi/2$, and it starts at 0. Figure 10.2 shows the levels against the strength γ.

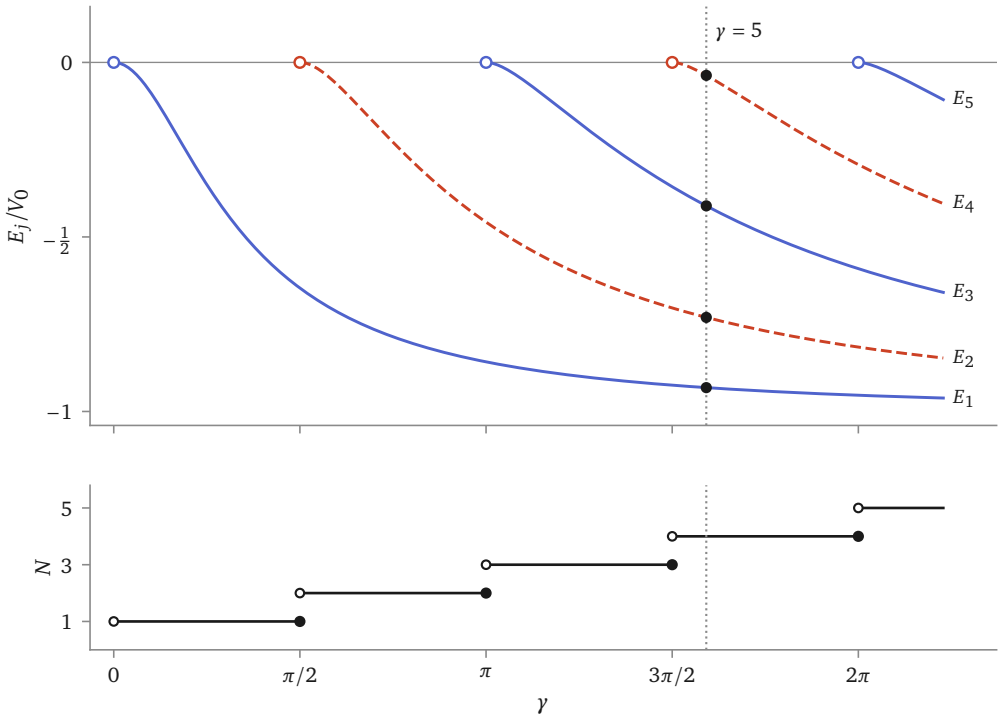


Figure 10.2: The levels of the square well against its strength γ, from the crossings of Theorem 10.8. Top: E_j/V_0 for $j = 1, \dots, 5$, solid for an even and dashed for an odd bound state. The level E_j exists for $\gamma > (j-1)\pi/2$. The open circle at $E = 0$ marks where it starts. That point is not a level. The dotted line at $\gamma = 5$ meets the four levels of Figure 10.1. Bottom: the number of levels $\lceil 2\gamma/\pi \rceil$. At $\gamma = n\pi/2$ it is still n (filled dot), and just above it is $n + 1$ (open circle).

So the bound states of a potential are the nonzero square-integrable solutions of (10.1), its levels are negative and simple, and the square well has exactly $\lceil 2\gamma/\pi \rceil$ levels, all in $(-V_0, 0)$.

Self-check 10.4

1. For which values of γ does the square well have exactly two levels? Is the bound state of E_2 even or odd?

Lecture 10, second part

The continuum, and why every attractive well binds

10.5 The continuum

The first part proved that the levels of a potential are negative and simple. It also found every level of the square well. In this part we prove that the essential spectrum is $[0,\infty)$, and that below 0 the spectrum consists of isolated simple eigenvalues. Then we prove that every potential with $\int V < 0$ has at least one level.

The proof of Theorem 10.10 begins with the packets of § 6.2, moved to the right of the well, where V vanishes. They show that every $\lambda \geq 0$ lies in the spectrum of H. Figure 10.3 draws them.

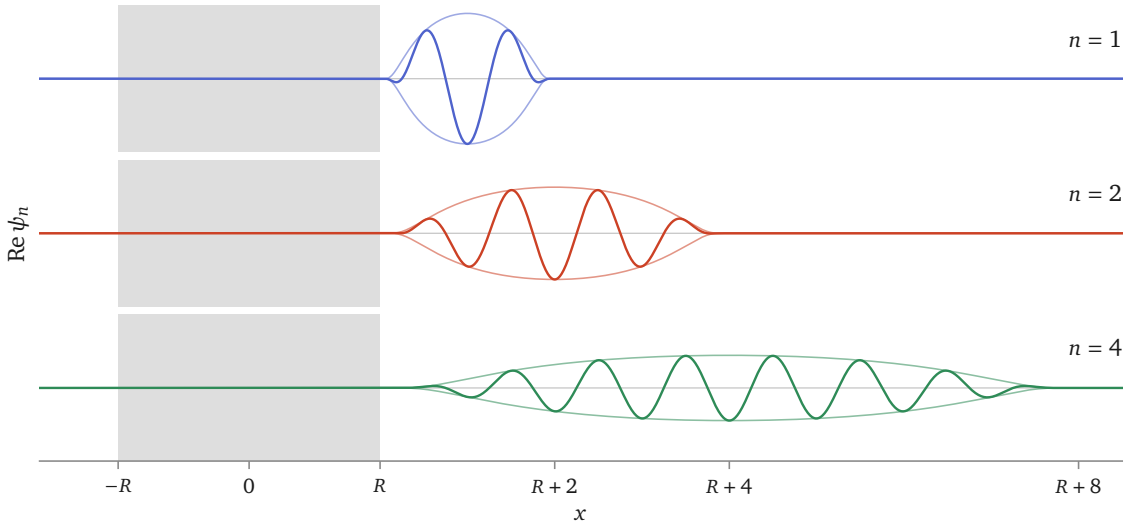


Figure 10.3: The packets $\psi_n(x) = n^{-1/2} f((x-c_n)/n) e^{ip_0 x/\hbar}$, with $c_n = R+n$, of the proof of Theorem 10.10, for $n = 1$, 2 and 4, on one vertical scale. Here $f(y) = Ce^{-1/(1-y^2)}$ for $|y| < 1$ and $f(y) = 0$ otherwise, with $C > 0$ such that $\|f\| = 1$. Lengths are in the unit in which f vanishes outside $[-1,1]$; $R = 1.5$ and $p_0 = 2\pi\hbar$, so the wavelength is 1. Each row shows $\operatorname{Re}\psi_n$ between $|\psi_n|$ and $-|\psi_n|$. The potential vanishes outside the shaded interval $[-R,R]$. The packet ψ_n vanishes outside $[R, R+2n]$, so $V\psi_n = 0$. As n grows, the wavelength $2\pi\hbar/p_0$ stays the same, the height falls like $n^{-1/2}$, and the centre c_n moves away from the well.

Above 0 the particle can leave the well, and the spectrum there is that of the free particle. Below 0 it cannot, and the spectrum is discrete. The second claim rests on one lemma: on a bounded interval, a sequence whose free energy stays bounded has a convergent subsequence. A potential acts only on a bounded interval. So for a Weyl sequence below 0, the lemma makes the vectors $V\psi_n$ converge along a subsequence, and the limit of the sequence is then an eigenvector.

Lemma 10.9 (Local compactness). *Let $\eta > 0$, $z = -\hbar^2\eta^2/2m$ and $R > 0$, and let $\psi_n \in D(H_0)$ with $\|(H_0 - z)\psi_n\| \leq C$ for every n. Then a subsequence of $(1_{[-R,R]}\psi_n)$ converges in norm.*

Proof. By the first claim of Lemma 10.4, with $\varphi = (H_0 - z)\psi_n$, each ψ_n is continuous, $|\psi_n(x)| \le C\,\|G\|$, and $|\psi_n(x) - \psi_n(x')| \le C\,\|G_x - G_{x'}\|$, which tends to 0 with $|x - x'|$ at the same rate for all n. Let $r_1, r_2, \dots$ be the rational points of $[-R, R]$. By the Bolzano–Weierstrass theorem, a subsequence converges at r_1, a subsequence of it at r_2, and so on. The diagonal sequence (θ_l), whose lth term is the lth term of the lth subsequence, converges at every r_i. Let $\epsilon > 0$. Choose $\delta > 0$ with $C\,\|G_x - G_{x'}\| < \epsilon$ for $|x - x'| < \delta$, finitely many r_i such that every point of $[-R, R]$ lies within δ of one of them, and L with $|\theta_l(r_i) - \theta_{l'}(r_i)| < \epsilon$ for $l, l' \ge L$ and these r_i. Then $|\theta_l(x) - \theta_{l'}(x)| < 3\epsilon$ for every $x \in [-R, R]$, so $\left\|1_{[-R,R]}(\theta_l - \theta_{l'})\right\|^2 \le 18R\epsilon^2$. The sequence $(1_{[-R,R]}\theta_l)$ is a Cauchy sequence, and it converges. ∎

This is the Arzelà–Ascoli argument. With it, the spectrum of H falls into two parts.

Theorem 10.10 (A spectrum in two parts)**.** *Let V be a potential. Then $\sigma_{\text{ess}}(H) = [0, \infty)$. Every point of $\sigma(H)$ below 0 is an eigenvalue in $\sigma_{\text{d}}(H)$, and these eigenvalues can accumulate only at 0.*

Proof. First, $[0, \infty) \subseteq \sigma(H)$. Let V vanish outside $[-R, R]$, let $\lambda = p_0^2/2m$ with $p_0 \ge 0$, let f be a real smooth function that vanishes outside $[-1, 1]$, with $\|f\| = 1$, and put

$$\psi_n(x) = n^{-1/2} f\Big(\frac{x - c_n}{n}\Big) e^{\mathrm{i}p_0 x/\hbar}, \qquad c_n = R + n.$$

These are the packets of § 6.2, moved to the right of the well, so $V\psi_n = 0$. By Theorem 4.6,

$$(H - \lambda)\psi_n = -\frac{\hbar^2}{2m}\, n^{-1/2}\Big(\frac{f''}{n^2} + \frac{2\mathrm{i}p_0}{\hbar n}\, f'\Big)\Big(\frac{x - c_n}{n}\Big) e^{\mathrm{i}p_0 x/\hbar},$$

and the substitution $x = c_n + ny$ gives

$$\|(H - \lambda)\psi_n\| \le \frac{\hbar^2}{2m}\Big(\frac{\|f''\|}{n^2} + \frac{2p_0\,\|f'\|}{\hbar n}\Big) \longrightarrow 0.$$

By Proposition 6.2, $\lambda \in \sigma(H)$. No point of the interval $[0, \infty)$ is isolated in $\sigma(H)$, so $[0, \infty) \subseteq \sigma_{\text{ess}}(H)$.

Next, let $\lambda < 0$ lie in $\sigma(H)$. By Weyl's criterion (Theorem 6.3) there is a Weyl sequence (ψ_n) for λ. Fix $z < 0$. Since $(H_0 - z)\psi_n = (H - \lambda)\psi_n + (\lambda - z)\psi_n - V\psi_n$,

$$\|(H_0 - z)\psi_n\| \le \|(H - \lambda)\psi_n\| + |\lambda - z| + \sup|V|$$

is bounded. By Lemma 10.9, for a subsequence $1_{[-R,R]}\psi_n$ converges, and so does $V\psi_n = V1_{[-R,R]}\psi_n$. Then $(H_0 - \lambda)\psi_n = (H - \lambda)\psi_n - V\psi_n$ converges to a vector χ. Since $\lambda < 0$ lies outside $\sigma(H_0) = [0, \infty)$ (Proposition 6.8), the resolvent $(H_0 - \lambda)^{-1}$ is bounded, and

$$\psi_n = (H_0 - \lambda)^{-1}(H_0 - \lambda)\psi_n \longrightarrow \psi = (H_0 - \lambda)^{-1}\chi,$$

a unit vector in $D(H_0)$. Since $V\psi_n \to V\psi$, the limit is $\chi = -V\psi$, so $(H_0 - \lambda)\psi = -V\psi$ and $H\psi = \lambda\psi$. By Theorem 10.7, λ is a simple eigenvalue.

Then, λ is isolated. Otherwise there are distinct points $\lambda_l \ne \lambda$ of $\sigma(H)$ with $\lambda_l \to \lambda$, negative for large l, so eigenvalues by the previous step, with unit eigenvectors ψ_l. Since H is self-adjoint, $(\lambda_l - \lambda_{l'})\,\langle\psi_l, \psi_{l'}\rangle = 0$, so the ψ_l are orthonormal, and $\|(H - \lambda)\psi_l\| = |\lambda_l - \lambda| \to 0$. By the previous step, a subsequence converges in norm, but $\|\psi_l - \psi_{l'}\|^2 = 2$ for $l \ne l'$. So $\lambda \in \sigma_{\text{d}}(H)$. Finally, a limit $\lambda < 0$ of negative eigenvalues would lie in the closed set $\sigma(H)$ and not be isolated, which we have excluded. ∎

In general, this is Weyl's theorem (Teschl, Thm 6.19): a perturbation that changes the resolvent by a compact operator keeps the essential spectrum. For the square well, by Theorems 10.8 and 10.10,

$$\sigma(H) = \{E_1, \dots, E_N\} \cup [0, \infty).$$

The levels are the discrete spectrum, and the continuum is the essential spectrum, with no eigenvalue in it (Theorem 10.7). Lecture 11 says what the continuum's states are.

Self-check 10.5

1. For the square well with $\gamma = 5$ of Figure 10.1, find $\sigma_{\mathrm{d}}(H)$ and $\sigma_{\mathrm{ess}}(H)$. Is 0 an eigenvalue?

10.6 Every attractive well binds

Every square well has a level. The reason is not its shape.

Theorem 10.11 (Every attractive well binds)**.** *Let V be a potential with $\int V \,\mathrm{d}x < 0$. Then the least point of $\sigma(H)$ is a negative eigenvalue of H.*

Proof. Let V vanish outside $[-R, R]$. Let f be a real smooth function that vanishes outside $[-2, 2]$ and equals 1 on $[-1, 1]$. For $0 < \epsilon \le 1/R$ put $\psi_\epsilon(x) = \sqrt{\epsilon}\, f(\epsilon x)$. Then $\psi_\epsilon = \sqrt{\epsilon}$ on $[-R, R]$, so

$$\langle \psi_\epsilon, V\psi_\epsilon \rangle = \epsilon \int V \,\mathrm{d}x.$$

By Theorem 4.6 and an integration by parts, $\langle \psi_\epsilon, H_0\psi_\epsilon \rangle = (\hbar^2/2m)\left\|\psi_\epsilon'\right\|^2$, and the substitution $y = \epsilon x$ gives $\left\|\psi_\epsilon'\right\|^2 = \epsilon^2 \left\|f'\right\|^2$ and $\|\psi_\epsilon\| = \|f\|$. So

$$\langle \psi_\epsilon, H\psi_\epsilon \rangle = \epsilon\Big(\int V \,\mathrm{d}x + \epsilon \,\frac{\hbar^2}{2m} \left\|f'\right\|^2 \Big),$$

which is negative for small ϵ. By Proposition 10.3, the infimum λ_0 of the means, which is at least $-\sup|V|$, is the least point of $\sigma(H)$, and the trial state $\psi_\epsilon/\|f\|$ shows that $\lambda_0 < 0$. By Theorem 10.10, λ_0 is an eigenvalue. ∎

A wide, low state ψ_ϵ costs a kinetic energy of order ϵ^2 and gains a potential energy of order ϵ, and in one dimension the gain wins. Figure 10.4 shows this for one square well.

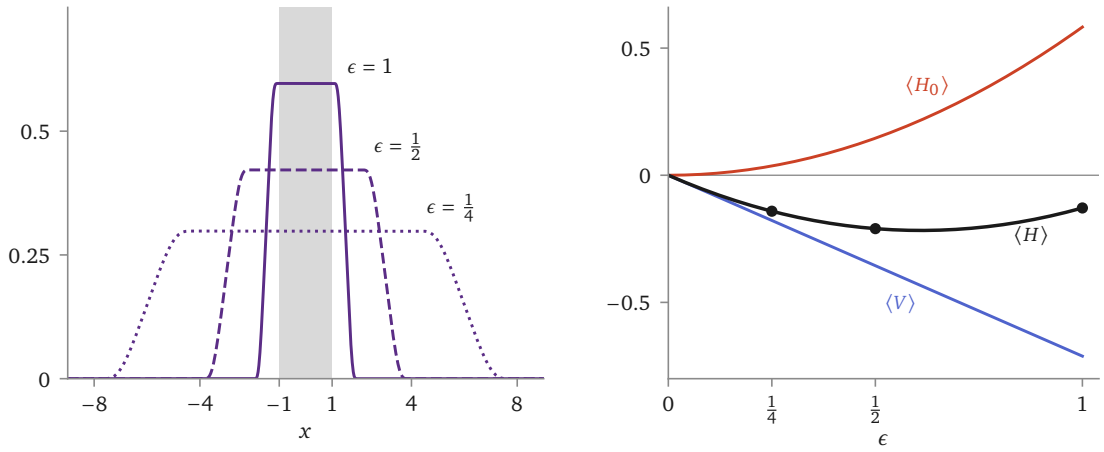


Figure 10.4: The trial states of the proof of Theorem 10.11 for the square well with $V_0 = \hbar^2/mw^2$, in units with $\hbar = m = w = 1$, so that $R = 1$. Here $f = 1$ on $[-1, 1]$, $f = 0$ outside $(-2, 2)$, and $f(x) = h(2 - |x|)/\big(h(2 - |x|) + h(|x| - 1)\big)$ in between, with $h(t) = e^{-1/t}$. Left: the unit trial states $\psi_\epsilon/\|f\|$ for $\epsilon = 1, \frac{1}{2}, \frac{1}{4}$, over the shaded well. Right: the means $\langle H_0 \rangle = \epsilon^2 \|f'\|^2 / 2\|f\|^2$ and $\langle V \rangle = -2\epsilon/\|f\|^2$ in these states, and their sum $\langle H \rangle$, for $0 < \epsilon \leq 1$, where the formula of the proof holds. The dots mark the three states on the left.

The converse is false: a potential with $\int V > 0$ can bind. Take $\hbar^2/mw^2$ as the unit of energy, the square well with $V_0 = 2$ (so $\gamma = 2$), and add a barrier of height 5 on $[6w, 7w]$. Then $\int V = -4w + 5w > 0$. Take as trial state the bound state ψ_1 of the well alone, normalised. It has $\langle \psi_1, H\psi_1 \rangle = E_1 + 5 \int_{6w}^{7w} |\psi_1|^2 \, dx \approx -1.47$, since ψ_1 decays like $e^{-\eta_1 x}$ beyond w. By Proposition 10.3 and Theorem 10.10, the new H has a negative eigenvalue. So $\int V < 0$ is sufficient for binding, and not necessary.

Self-check 10.6

1. Where does the proof of Theorem 10.11 fail if $\int V \, dx = 0$?

The physics

In 1974 Dingle, Wiegmann and Henry grew layers of GaAs, the thinnest about 7 nm thick, between layers of AlGaAs, in which an electron has a higher energy. Light absorbed in such a layer showed up to eight resolved lines, from the levels of the electrons and of the holes together, and they described the layer as a simple rectangular well for each. In a model with a constant effective mass, an electron near the bottom of the GaAs band at low temperature moves as a particle of mass $0.067\, m_e$. For a layer 10 nm thick ($w = 5$ nm) and a depth $V_0 = 0.3$ eV, chosen as an example, $\gamma \approx 3.63$, so the well has $\lceil 2.31 \rceil = 3$ levels, at about 34, 133 and 272 meV above its bottom. Changing the thickness shifts the levels and can change their number.

The finite well

States: the bound states are the square-integrable, continuously differentiable solutions of (10.1) (Proposition 10.6); for a potential V (Definition 10.1), they decay like $e^{-\eta|x|}$.

A square well has $\lceil 2\gamma/\pi \rceil$ levels (Theorem 10.8), and every such V with $\int V < 0$ has at least one (Theorem 10.11). *Observables:* for such V, $H = H_0 + V$ is self-adjoint on $D(H_0)$ (Proposition 10.2). Its essential spectrum is $[0, \infty)$, and below 0 its spectrum consists of isolated simple eigenvalues (Theorems 10.7 and 10.10). *Measurement:* by the cited Theorem 6.4, an energy reading of a particle in a square well lies in $\{E_1, \dots, E_N\} \cup [0, \infty)$.

Summary.

- *A bounded potential keeps self-adjointness* (Proposition 10.2), and for a self-adjoint operator whose means are bounded below, *the bottom of the spectrum* is the infimum of the means, which is the variational principle (Proposition 10.3).
- *The eigenvalue equation*: the eigenvectors of H are the square-integrable, continuously differentiable solutions of (10.1) (Proposition 10.6); for a potential that vanishes outside a bounded interval, every level is negative and simple (Theorem 10.7).
- *The square well* has $\lceil 2\gamma/\pi \rceil$ levels, alternately even and odd (Theorem 10.8).
- *A spectrum in two parts*: $\sigma_{\text{ess}}(H) = [0, \infty)$, and the spectrum below 0 is discrete (Theorem 10.10); *every attractive well binds* if $\int V < 0$ (Theorem 10.11).

Lecture 11 asks what the states of the continuum are.

Exercises. Exercise Sheet 10.

Self-check answers.

10.1 1. For a unit $\psi \in D(H)$, $\langle \psi, H\psi \rangle = \langle \psi, H_0\psi \rangle + \langle \psi, V\psi \rangle \geq 0 - V_0$, so $\lambda_0 \geq -V_0$, and Proposition 10.3 gives $\sigma(H) \subseteq [\lambda_0, \infty) \subseteq [-V_0, \infty)$.

10.2 1. For $x > 0$, $\psi'' = \eta^2\psi$, and the same for $x < 0$. But $\psi'(0+) - \psi'(0-) = -2\eta$, so ψ is not continuously differentiable, and by Proposition 10.6 with $V = 0$ it is not an eigenvector. Indeed H_0 has no eigenvalue (Theorem 4.6).

10.3 1. Since V and E are real, the complex conjugate of (10.1) is the same equation for $\overline{\psi}$, so $\overline{\psi}$ is a bound state for E (Proposition 10.6). By Theorem 10.7, $\overline{\psi} = c\psi$, with $|c| = 1$. Write $c = e^{2i\theta}$. Then $\overline{e^{i\theta}\psi} = e^{-i\theta}c\psi = e^{i\theta}\psi$.

10.4 1. Exactly when $\lceil 2\gamma/\pi \rceil = 2$, that is, $\pi/2 < \gamma \leq \pi$. The bound state of E_2 is odd.

10.5 1. $\sigma_{\text{d}}(H) = \{E_1, E_2, E_3, E_4\}$ and $\sigma_{\text{ess}}(H) = [0, \infty)$. By Theorem 10.7, 0 is not an eigenvalue.

10.6 1. Then $\langle \psi_\epsilon, H\psi_\epsilon \rangle = \epsilon^2(\hbar^2/2m)\,\|f'\|^2 > 0$ for every ϵ, so the proof gives no negative upper bound. It decides nothing about such a V.

References

Other books prove some results of this lecture. Teschl, Thm 2.19, states Proposition 10.3.

Teschl, *Mathematical Methods in Quantum Mechanics*, Thm 2.19 (p. 77), Thm 6.4 (p. 135), Thm 6.19 (p. 146). *Linear Algebra for Physics*, Thm 7.12. Dingle, Wiegmann and Henry, *Phys. Rev. Lett.* 33, 827 (1974). Vurgaftman, Meyer and Ram-Mohan, *J. Appl. Phys.* 89, 5815 (2001).